SDG BLOCKCHAIN ACCELERATOR

Debugging and Testing Report

1. Project Information

- **Project Name:** Digital Infrastructure for Green Energy Exports / Fuel Switch x UNDP BiH Pilot - Cohort 2
- **Challenge & UNDP Office:** Bosnia & Herzegovina Country Office

- **Report Version:** 1

2. Testing Approach

Unit Testing:

- Tool: Aiken CLI (aiken check).
- Coverage: 8 scenario-based tests covering valid batch minting, burn integrity, and security failures.
- Edge Cases: Tested kw values exceeding the 1000 hard-cap, asset name mismatches (spoofing), and illegal positive mints during a burn action.

Integration Testing:

- Environment: Conducted on the Cardano Preprod Testnet.
- Scope: End-to-end verification of the Node.js API, Redis idempotency layer, and on-chain validator execution via the Lucid library.

Edge Cases:

- Double-Mint Protection: Verification that identical site/date/energy requests are blocked by the Redis cache.
- UTXO Contention: Testing "Fan-Out" logic to ensure the system can handle concurrent mints without "Busy" errors

3. Error Logs

- Error 1 (Aiken CLI Fail): [FAIL] mint_02_fail_kw_too_high — Expected: Validation failure

  Got: Successfully blocked 1001 kWh mint.

- Error 2 (API Idempotency):

POST /mint -> 409 Conflict: "Double Mint Detected" Logged when Redis identifies an existing Batch ID for a site.

- Error 3 (Auth Failure):

GET /mint -> 401 Unauthorized — Logged when a request is made without the x-api-key header

4. Resolved Issues

Issue ID	*Description*	*Root Cause*	*Resolution*	*Status*
001	*Duplicate Asset Names*	*Site/Date combo was not unique for multiple batches.*	*Added batch_id to RecDetails and name construction.*	*Fixed*
002	*UTXO Contention*	*High-throughput minting led to "UTXO Already Spent" errors.*	*Implemented "Fan-Out" strategy to split large UTXOs into 3+ pure ADA chunks.*	*Fixed*
003	*Energy Inflation*	*Potential for typos to mint massive unverified energy values.*	*Enforced kw <= 1000 hard-cap in the Aiken smart contract.*	*Fixed*

Issue ID	*Description*	*Root Cause*	*Resolution*	*Status*
004	*Metadata Spoofing*	*Off-chain server could theoretically inject false site data.*	*Deterministic on-chain name reconstruction from redeemer metadata.*	*Fixed*

5. Optimization Notes

- Validator Refactoring: Flattened batch validation using list.all in Aiken to iterate through multi-asset mints efficiently.
- Plutus V3 Utilization: Compiled specifically for Plutus V3 to leverage lower execution costs and improved script efficiency.
- Memory Management: Implemented UTXO masking in the backend to filter out non-essential inputs, preventing "Collateral" errors during transaction building.

6. Tools and Environments Used

- Aiken CLI: v1.1.19.
- Cardano Network: Preprod Testnet.
- Backend Stack: Node.js (Express), Redis (ioredis), Blockfrost API, and @lucid evolution/lucid .
- Deployment: Render

7. Remaining Issues / Next Steps

- Architectural Pivot: Transition from individual NFTs to Fungible Tokens (FTs) to allow bulk minting in a single transaction, significantly reducing fees.

- Persistence Tuning: Configure Redis with AOF (Append Only File) persistence to ensure the idempotency cache is not lost during server restarts.
- Performance: Optimize the "Transaction Chunking" loop to maximize the limit of 15 assets per transaction under varied network conditions.